Cambridge (CIE) A Level Maths: Pure 3

Your notes

Laws of Logarithms

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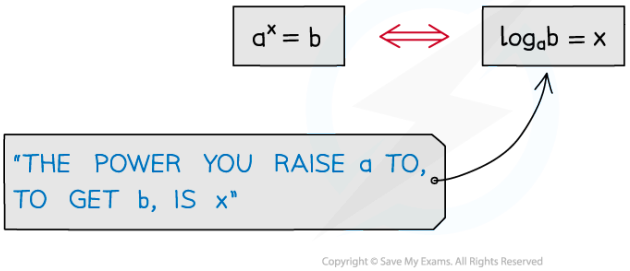
Your notes

Laws of logarithms

What are the laws of logarithms?



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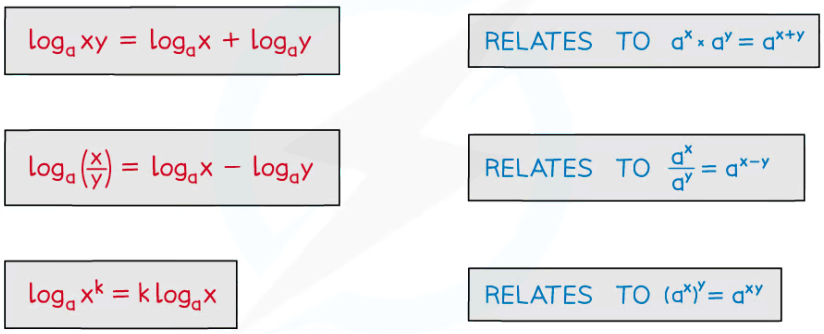
- There are many laws or rules of indices, for example
  - $a^m \times a^n = a^{m+n}$
  - $(a^m)^n = a^{mn}$
- There are equivalent laws of logarithms (for a > 0)
  - $\log_a xy = \log_a x + \log_a y$
  - $\log_a \left(\frac{x}{y}\right) = \log_a x - \log_a y$
  - $\log_a x^k = k \log_a x$



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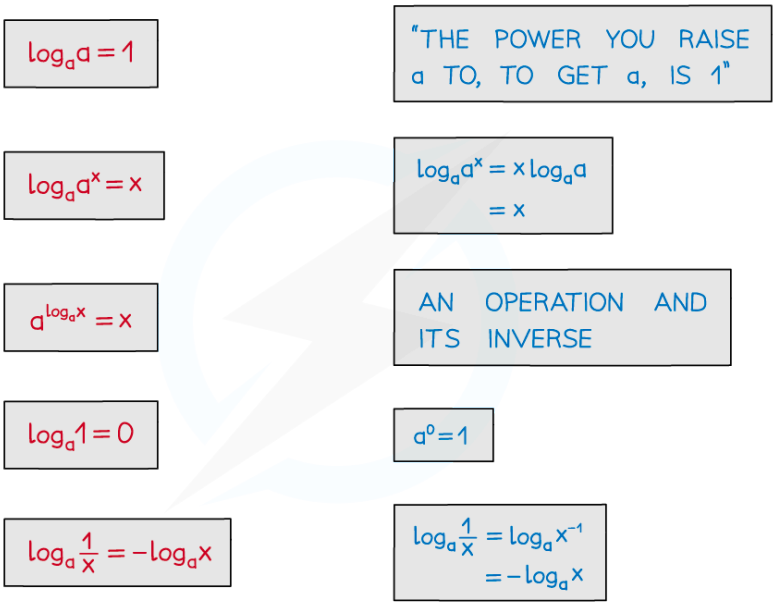
- There are also some particular results these lead to
  - $\log_a a = 1$
  - $\log_a a^x = x$
  - $a^{\log_a x} = x$
  - $\log_a 1 = 0$
  - $\log_a\left(\frac{1}{x}\right) = -\log_a x$



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- Two of these were seen in the notes Logarithmic Functions
- Beware ...
  - ... **log (x + y) ≠ log x + log y**
- Results apply to **ln** too
  - $\ln x \equiv \log_e x$
  - In particular $e^{\ln x} = x$ and $\ln(e^x) = x$

How do I use log laws?

- Laws of logarithms can be used to ...

- ... simplify expressions
- ... solve logarithmic equations
- ... solve exponential equations

e.g. WRITE $3\log_2(2x+3) + \log_2 5 - 2\log_2(x+1)$ AS A SINGLE LOGARITHM

$$3\log_2(2x+3) + \log_2 5 - 2\log_2(x+1) = \log_2(2x+3)^3 + \log_2 5 - \log_2(x+1)^2$$

$k\log_a x = \log_a x^k$

$$= \log_2 5(2x+3)^3 - \log_2(x+1)^2$$

$\log_a x + \log_a y = \log_a(xy)$

$$= \log_2 \frac{5(2x+3)^3}{(x+1)^2}$$

$\log_a x - \log_a y = \log_a \frac{x}{y}$

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Examiner Tips and Tricks

- Remember to check whether your solutions are valid. log (x+k) is only defined if x > -k.
- You will lose marks if you forget to reject invalid solutions.

Worked Example

Your notes

Your notes

Solve the equation $2\log_2(x) = 3 + \log_2(x+6)$

$\log_2(x^2) = 3 + \log_2(x+6)$

$k\log_a x = \log_a x^k$

$\log_2(x^2) - \log_2(x+6) = 3$

LOGARITHMS ON THE SAME SIDE

$\log_2\left(\frac{x^2}{x+6}\right) = 3$

$\log_a x - \log_a y = \log_a\left(\frac{x}{y}\right)$

$\frac{x^2}{x+6} = 2^3$

$\log_a x = y \Leftrightarrow x = a^y$

$x^2 = 8(x+6)$

$x^2 - 8x - 48 = 0$

$x = -4$ or $x = 12$

FORM AND SOLVE A QUADRATIC

CHECK WHETHER SOLUTIONS ARE VALID $\log_a x$ IS ONLY DEFINED WHEN $x > 0$

$\log_2 x \Rightarrow x > 0$

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$\log_2(x+6) \Rightarrow x + 6 > 0 \Rightarrow x > -6$

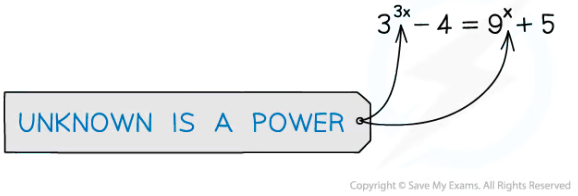
REJECT $x = -4$

$\therefore x = 12$

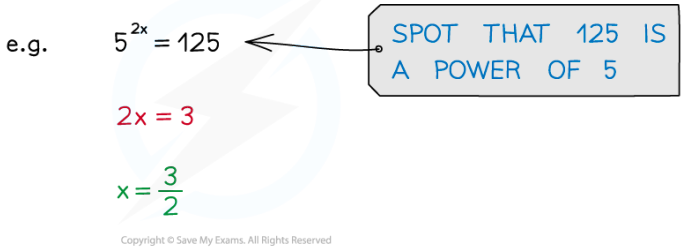
Your notes

Exponential equations

What are exponential equations?



- An equation where the unknown is a power
- In simple cases the solutions can be "spotted"



How do I solve exponential equations?

- Laws of Indices are often needed to rewrite equations
- Laws of Logarithms are used to solve exponential equations
- **ln** (**$\log_e$**) is often used
- Answers are often written in terms of **ln**

Your notes

e.g. SOLVE $3^{2x-1} = 2^{6-x}$ GIVING YOUR ANSWER IN THE FORM $x = \frac{\ln p}{\ln q}$ WHERE p AND q ARE INTEGERS TO BE FOUND

STEP 1: TAKE LOGARITHMS OF BOTH SIDES

$$\ln 3^{2x-1} = \ln 2^{6-x}$$

STEP 2: USE LAWS OF LOGARITHMS TO REMOVE POWERS

$$(2x-1)\ln 3 = (6-x)\ln 2$$

STEP 3: REARRANGE TO ISOLATE x

$$2x\ln 3 - \ln 3 = 6\ln 2 - x\ln 2$$

$$2x\ln 3 + x\ln 2 = 6\ln 2 + \ln 3$$

$$x(2\ln 3 + \ln 2) = 6\ln 2 + \ln 3$$

$$x = \frac{6\ln 2 + \ln 3}{2\ln 3 + \ln 2}$$

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STEP 4: USE LAWS OF LOGARITHMS TO REWRITE IN CORRECT FORM

$$x = \frac{\ln 2^6 + \ln 3}{\ln 3^2 + \ln 2}$$

$$x = \frac{\ln 64 + \ln 3}{\ln 9 + \ln 2}$$

$$x = \frac{\ln 192}{\ln 18}$$

$p = 192$
$q = 18$

$\log_a x + \log_a y = \log_a xy$

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Your notes

e.g. SOLVE $6^{3x+2} = 42$, GIVING YOUR ANSWER CORRECT TO 3 SIGNIFICANT FIGURES

$$6^{3x+2} = 42$$

$$\ln(6^{3x+2}) = \ln 42$$

TAKE LOGARITHMS OF BOTH SIDES

$$(3x+2)\ln 6 = \ln 42$$

$$3x+2 = \frac{\ln 42}{\ln 6}$$

$\neq \ln 7$

$$3x = \frac{\ln 42}{\ln 6} - 2$$

$$x = \frac{\frac{\ln 42}{\ln 6} - 2}{3}$$

$$x = 0.0286777...$$

AVOID USING CALCULATOR UNTIL NECESSARY

$$x = 0.0287 \quad (3\text{ sf})$$

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How do I solve hidden quadratic exponential equations?

- Look out for "hidden" squared terms, these are hidden quadratics which will need to be solved
  - $4^x = (2^2)^x = 2^{2x} = (2^x)^2$
  - $e^{2x} = (e^2)^x = (e^x)^2$

Your notes

e.g. SOLVE $21e^x - 4 = 5e^{2x}$ GIVING YOUR SOLUTIONS IN THE FORM $x = a\ln b$, WHERE a AND b ARE INTEGERS

"HIDDEN" SQUARED TERM

$21e^x - 4 = 5e^{2x}$

$5(e^x)^2 - 21(e^x) + 4 = 0$

REARRANGE TO QUADRATIC FORM

$(5e^x - 1)(e^x - 4) = 0$

USE $y = e^x$ IF PREFERRED

$e^x = \frac{1}{5}$ $\quad$ $e^x = 4$

$x = \ln\frac{1}{5}$ $\quad$ $x = \ln 4$

$x = \ln 5^{-1}$

REWRITE IN CORRECT FORM

$x = -\ln 5$ $\quad$ $x = \ln 4$

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Worked Example

Your notes

Find the exact solution to $6 \times 3^{2x-3} - 2^{x+1} = 0$

$6 \times 3^{2x-3} = 2^{x+1}$

$\ln(6 \times 3^{2x-3}) = \ln 2^{x+1}$

STEP 1: LOGS OF BOTH SIDES

$\ln 6 + \ln 3^{2x-3} = (x+1)\ln 2$

$\ln 6 + (2x-3)\ln 3 = (x+1)\ln 2$

STEP 2: REMOVE POWERS

$\ln 6 + 2x\ln 3 - 3\ln 3 = x\ln 2 + \ln 2$

$2x\ln 3 - x\ln 2 = \ln 2 - \ln 6 + 3\ln 3$

$x(2\ln 3 - \ln 2) = \ln 2 - \ln 6 + 3\ln 3$

$$x = \frac{\ln\frac{2}{6} + \ln 3^3}{\ln 3^2 - \ln 2}$$

STEP 3: ISOLATE x

$$x = \frac{\ln\frac{1}{3} + \ln 27}{\ln 9 - \ln 2}$$

STEP 4: REWRITE USING LAWS OF LOGARITHMS

$$x = \frac{\ln 9}{\ln\frac{9}{2}}$$

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